

Interpretable NSFW Text Moderation via RISE and Post-hoc Concept Bottleneck Models

Raymond Kang
Yunhan Wei

1. Introduction / Motivation

Online platforms moderate billions of text posts and comments per day. Scaling this work safely requires automated classifiers that can detect Not Safe For Work (NSFW) content—including sexual content, hate speech, and graphic violence—with high accuracy, low latency, and predictable behavior. However, purely black-box models are difficult to audit, debug, or defend when users appeal a decision or when regulators require explanations. In this project, we propose an interpretable moderation pipeline that marries perturbation-based attribution (RISE) with post-hoc Concept Bottleneck Models (CBMs) to provide multi-level, human-readable explanations for text moderation outcomes.

Concretely, we target three objectives: (i) preserve competitive classification performance on NSFW detection tasks; (ii) expose concept-level rationales (e.g., ‘sexual innuendo’, ‘targeted insult’, ‘implicit threat’) that sit between tokens and the final decision; and (iii) provide usable visual analytics that help moderators and developers understand model failures, reduce bias, and iterate quickly. We hypothesize that a RISE+CBM stack will yield more faithful and more useful explanations than token-only saliency while requiring no retraining of the base model.

Key challenges include: preventing spurious correlations in concept extraction; ensuring explanation faithfulness; avoiding exposure of sensitive content in visualizations; and meeting interactive latency budgets. Our technical approach introduces a lightweight concept projection trained post-hoc on hidden states, a text-adapted RISE that perturbs subword tokens, and a co-designed visualization that links token saliency, concept activations, and final labels.

2. Background and Related Work

2.1 Token-/Feature-level Explanations (LIME, SHAP, IG, Attention)

Early model-agnostic explanation methods include LIME, which fits a simple surrogate around each prediction via local perturbations, and SHAP, which frames feature attributions as Shapley values with desirable axiomatic properties. LIME is simple and flexible but can be sensitive to sampling; SHAP provides theoretical guarantees but can be costly in high dimensions. Integrated Gradients (IG) attributes prediction changes along a path from a baseline to the input and is widely used for Transformer text classifiers; it comes with axiomatic guarantees (Sensitivity, Implementation Invariance). Whether attention is an explanation has been debated: Attention is not Explanation argues attention weights often don’t align with causal importance; Attention is Explanation counters that, under certain tests and definitions, attention can still provide insight. In our work, we treat attention as supporting context and use perturbation-based methods for primary evidence.

2.2 Concept-Level Interpretability (TCAV, Concept Bottleneck Models)

[TCAV](#) connects internal representations to human concepts via “concept activation vectors,” quantifying how sensitive predictions are to high-level concepts (e.g., insult, sexual innuendo). Concept Bottleneck Models (CBM) explicitly insert a “concept layer” so the overall mapping is input $\rightarrow$ concepts $\rightarrow$ label, enabling interventions (edit concepts, observe label changes). We follow a post-hoc CBM

strategy—freeze the backbone classifier and train lightweight concept heads—balancing interpretability with minimal disruption to the base model.

2.3 RISE and Perturbation-based Explanations for Text

RISE (Randomized Input Sampling) explains black-box models by applying random masks to inputs, observing score changes, and aggregating to a saliency map. Originally for images, the same principle carries to text by randomly masking tokens/subwords and measuring the effect on the target score; RISE popularized the deletion/insertion evaluation curves we will use as faithfulness metrics.

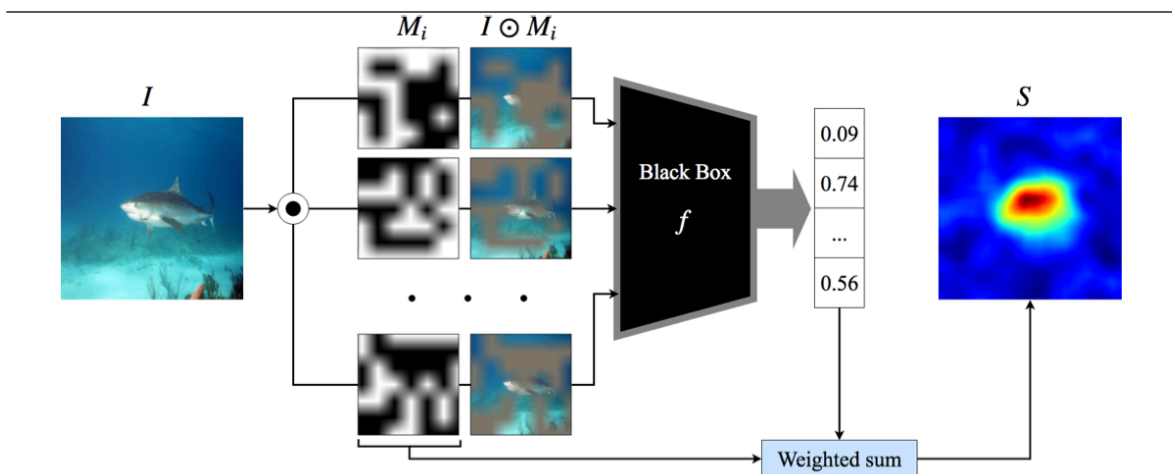


Figure1: RISE Pipeline: Randomized Input Sampling for Explaining Black-Box Models

2.4 Datasets

For toxicity and hate-speech moderation we will combine: [Civil Comments \(HF dataset card\)](#), [Jigsaw Unintended Bias](#), and [HateXplain GitHub](#).

3. Methodology

3.1 Structure

Our pipeline will include four modules:

- (A) a base Transformer classifier (outputs NSFW class scores);
- (B) a post-hoc concept bottleneck layer (projects hidden states into a human-readable concept vector);
- (C) a RISE-based text explainer (derives token importance via occlusion sampling);
- (D) a visualization dashboard (integrates token saliency heatmaps, concept bar charts, token↔concept associations, and a final decision card).

The overarching principle is to keep the base model unchanged while adding interpretability and room for intervention on top of it.

3.2 Dataset and Preparation

We will use the datasets referenced in 2.5. For efficiency and practical feasibility, we will unify formats across data sources and partition content to support training and final evaluation. We may also need to address class imbalance introduced by content stratification, for example through sample weighting.

3.3 Base Model Training

We will use RoBERTa-base as our base model. To handle potential issues, we may fine-tune the model to improve performance on the dataset and address interpretability concerns.

After the initial training, we will freeze the backbone weights to preserve the learned decision boundary, ensuring that our explanations remain strictly post-hoc.

3.4 Post-hoc Concept Bottleneck Module

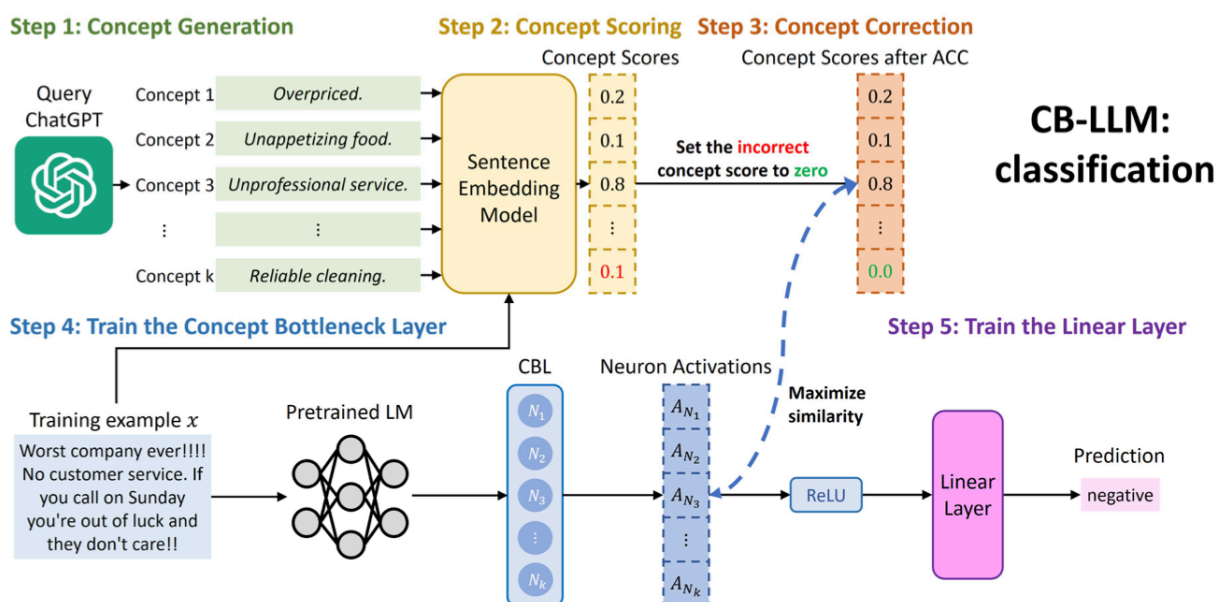


Figure 2. Other structural diagrams for concept-bottleneck large language models.

After training and freezing the base classifier, we attach a lightweight concept head that maps its hidden representations to 10–20 interpretable concept scores (0–1) relevant to NSFW moderation—such as sexual content, hate speech, threats, profanity, and self-harm. Each score indicates the strength of that concept in the text, allowing reviewers to see not only the output label but the reasoning behind it. Figure 2 will depict this as a bipartite graph linking influential words to activated concepts, with line thickness or color denoting contribution strength.

Concept supervision comes from three sources: (1) **Weak lexicon rules** using Hatebase, profanity, and sexual-slang lists refined by context filters; (2) **A small human-labeled subset** (hundreds of examples) to anchor and refine concept definitions; and (3) **Teacher models or APIs** that provide pseudo-labels for attributes like insult or threat. Together, these sources balance breadth, accuracy, and nuance.

We train a sparse multi-label classifier on frozen features so each concept depends on a few salient signals, enforcing intuitive directionality (e.g., more hate or threats can only increase NSFW likelihood). At inference, the concept vector is fused with the base model’s logits in a small scorer—preserving accuracy while adding interpretable evidence directly used for explanations.

Evaluation includes per-concept precision, recall, and F1 (target ≈ 0.6 F1), plus tests for concept independence from the overall label (e.g., profanity can appear in benign text). We further validate causal alignment by deletion/insertion checks—removing top highlighted words should sharply reduce the NSFW score, while adding them back should raise it. These ensure the concepts provide faithful, human-readable rationales suitable for moderation workflows.

3.5 RISE for Text Explanations

RISE for Text. For a sequence with length T , we create M random binary masks. Each mask decides which tokens to keep and which to hide, keeping a token with probability p and hiding it with probability $1-p$. Any hidden token is replaced with a neutral placeholder (for example, [MASK] in BERT or an OOV/pad/neutral embedding in RoBERTa).

For every masked version of the input, we pass it through the frozen base model and record the model’s score for the target class.

Token importance. We combine the scores from the masked inputs to estimate how important each token is. Tokens that appear more often in masks that receive high model scores are considered more influential. To do this, we create an importance vector I of length T and initialize it to zero. For each mask, we add the model score s_m to all positions in I where the mask has a value of 1 (meaning the token was kept). After processing all masks, we average and normalize the importance values. We also experiment with using low-discrepancy or orthogonal mask sampling to reduce variance, as well as span-based masking to better match natural language structure.

Efficiency. To support interactive use ($< \sim 0.5s/\text{input}$), we batch masked inputs on GPU, reuse computations with the frozen encoder (caching hidden states where possible), and leverage attention masks or baseline embeddings to approximate token removal. We cap MMM (e.g., $\sim 1k$) and use early stopping when rankings stabilize.

Validation & Figure. Faithfulness is measured via deletion/insertion curves: removing top-ranked tokens should sharply drop the NSFW score; inserting them should raise it. We compare against attention and Integrated Gradients and expect larger deletion AUC drops for RISE. Figure 3 will plot these curves (à la Petsiuk et al., 2018), highlighting RISE’s stronger alignment with model behavior.

3.6 Visualization

A web dashboard made from streamlit merges explanations:

- (i) Token Saliency Heatmap – highlights words by RISE importance;
- (ii) Concept Bar Chart – shows concept activations (e.g., Insult 0.9, Threat 0.2);
- (iii) Token–Concept Matrix – links influential tokens to activated concepts

We will use an Observable notebook to build the final results dashboard covering the elements above. For each text sample, we'll generate an object containing: sample ID, raw text, tokens, per-token labels and scores, token-level importance with respect to the target class, and the list of concepts. Having these elements we should easily generate figures we want to show results.

4. Study Design and Expected Result

Study Design

We will evaluate the explanation approach in three steps: a controlled comparison, case studies, and a small user study. The goal is to keep the base model fast and accurate while making explanations genuinely understandable and useful for debugging.

1) Controlled comparison

Using the same dataset splits, we compare three explanation conditions: token-only (Integrated Gradients), token-only (RISE), and our combined RISE + Concept Bottleneck (CBM) explanations. For model performance, we report standard metrics such as accuracy (and other common metrics as appropriate). For explanation quality, we run deletion and insertion tests—remove the top-ranked words and check whether the score drops sharply, then start from an empty input and add those words back to see whether the score rises quickly.

2) Case studies

We select real-world tricky examples, such as sentences that merely mention an identity without hate, slang or double entendre, English–Chinese code-switching, and sarcasm or irony. For each case, we first use RISE + CBM to pinpoint what's going wrong (e.g., the model over-reacts to an identity term), then apply a targeted fix (e.g., symmetric/counterfactual augmentation, reweighting, or refining a concept definition), and finally compare before vs. after to see whether errors decrease and explanations align better with human intuition. This shows how the tool actually helps find problems and improve the model.

3) User study

We recruit participants (e.g., classmates or colleagues) to judge model outputs. As in the controlled comparison, they see the three explanation conditions and then rate each on clarity, actionability, trust, and perceived faithfulness. We record their responses and analyze the results.

Expected study results

If the approach works as intended, we expect to observe:

1. Base accuracy remains essentially unchanged; RISE shows better faithfulness than attention on deletion/insertion curves and is comparable to IG, while being model-agnostic (no architectural access required).
2. Adding the CBM concept layer makes it easier for users to see why something was flagged and what to do next, leading to higher subjective ratings.
3. The concept head achieves solid recognition quality on the human-labeled subset (e.g., macro-F1 at or above our practical threshold), avoids “lighting up everything,” and does not simply mirror overall toxicity.

4. In bias-prone scenarios (e.g., mere identity mentions), explanations make the degree and source of bias visible; targeted fixes reduce false positives for affected groups.
5. Interaction is fast enough for practical use in a real tool.

Risks and limitations

If concept definitions are too broad or weak labels are too noisy, the concept head's interpretability and discriminability may degrade. For very long inputs or multi-turn context, perturbation-based explanations can become expensive and less stable. We plan mitigations such as a small, high-quality labeled subset to calibrate weak labels, clearer operational definitions for key concepts, segmentation and adaptive sampling for long texts, and front-end safeguards (e.g., masking sensitive terms and showing confidence cues).

5. Results

After training RoBERTa on 20,000 training examples (and 2000 validation examples) from each dataset (hatexplain, jigsaw, civil comments) we finished training our base classification model with a final training loss of 0.089 and a final validation accuracy of 0.8705. We then trained the concept bottleneck model (layer) and the final training loss was 0.0549. The accuracy is not amazingly high but due to limited computational resources and the scope of the project being about the visualization/explainability aspect we judged that this is satisfactory.

```
[Epoch 2/2] Step 2400/2500 Global 4900 Loss 0.0194 AvgLoss 0.0902
[Epoch 2/2] Step 2450/2500 Global 4950 Loss 0.0038 AvgLoss 0.0898
[Epoch 2/2] Step 2500/2500 Global 5000 Loss 0.1485 AvgLoss 0.0890
[Epoch 2] Training loss: 0.0890
[Epoch 2] Val acc: 0.8705
[CHECKPOINT] Saved base model checkpoint: checkpoints/base_model_epoch_2
[CHECKPOINT] Saved final base model to: checkpoints/base_model

[Concept Epoch 5] Training loss: 0.0549
[CHECKPOINT] Saved concept head checkpoint: checkpoints/concept_head_epoch_5.pt
[CHECKPOINT] Saved final concept head to: checkpoints/concept_head.pt
```

Figure 3: Images from the training of the RoBERTa base classification model and CBM model

After the training of the model, we tested the concept bottleneck model and the RISE based text explainers on various types of text categorized under the following categories (NSFW, not NSFW and uncertain phrases). We found that our model was able to mostly identify if the phrases were NSFW or not but also had some edge cases where a phrase was clearly not NSFW but the model labeled it as NSFW (example 3 fig 4). This is most likely due to the fact that some words may be used in NSFW and non-NSFW phrases and have differing meanings depending on the context. This is most likely due to the limited training examples and would most likely be solved if the model was trained for more epochs on more examples.


```

(.venv) raymond kang@Raymonds-MacBook-Air-4 newone % python3 demo_inference.py
Input: I hate you and your disgusting friends.
NSFW score: 0.9845
Concept scores: [0.99627006 0.48837605 0.49481532 0.44744578 0.41562155 0.54681927
0.4921225 0.5648415 0.48465127 0.53473806]
Some weights of RobertaForSequenceClassification were not initialized from the model checkpoint at distilroberta-
, 'classifier.dense.weight', 'classifier.out_proj.bias', 'classifier.out_proj.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

RISE NSFW score (original): 0.9845
Top tokens by importance:
<s> | importance=0.5307
</s> | importance=0.5307
Gdisgusting | importance=0.4712
Gyour | importance=0.3150
Ghate | importance=0.2977
. | importance=0.2834
Gfriends | importance=0.2709
Gand | importance=0.2655
Gyou | importance=0.2500
I | importance=0.2303

(.venv) raymond kang@Raymonds-MacBook-Air-4 newone % python3 demo_inference.py
Input: Hi my name is Jane.
NSFW score: 0.0009
Concept scores: [4.4325652e-04 5.3186017e-01 4.7203788e-01 3.7069052e-01 5.1413608e-01
4.6774879e-01 4.7466806e-01 3.4162852e-01 6.1171502e-01 4.6569496e-01]
Some weights of RobertaForSequenceClassification were not initialized from the model check
, 'classifier.dense.weight', 'classifier.out_proj.bias', 'classifier.out_proj.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predic

RISE NSFW score (original): 0.0009
Top tokens by importance:
<s> | importance=0.0023
</s> | importance=0.0023
GJane | importance=0.0010
. | importance=0.0009
Gmy | importance=0.0009
Hi | importance=0.0008
Gname | importance=0.0007
Gis | importance=0.0007

(.venv) raymond kang@Raymonds-MacBook-Air-4 newone % python3 demo_inference.py
Input: The quick brown fox jumps over the lazy dog.
NSFW score: 0.5962
Concept scores: [0.6784629 0.5254327 0.50154555 0.38661334 0.43112296 0.52326745
0.47959128 0.5829822 0.4684064 0.48898858]
Some weights of RobertaForSequenceClassification were not initialized from the model checkpoint at dist
, 'classifier.dense.weight', 'classifier.out_proj.bias', 'classifier.out_proj.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inf

RISE NSFW score (original): 0.5962
Top tokens by importance:
<s> | importance=0.4001
</s> | importance=0.4001
Glazy | importance=0.3640
Gfox | importance=0.2272
Gdog | importance=0.2207
The | importance=0.2028
. | importance=0.2013
Gover | importance=0.1898
Gthe | importance=0.1834
Gbrown | importance=0.1823

```

Figure 4: Manual testing of phrases using our trained model, RISE and CBM. Example 1 is NSFW, Ex 2 is not NSFW and Ex 3 is an example of a phrase that is not NSFW but our model labeled as NSFW

Visualization Results

We used streamlit to visualize the results where the number is RISE masks and the token keep probability can be adjusted. The visualization consists of an user inputting a phrase into a text box, then the phrase is used as input into our trained RoBERTa model with Concept Bottleneck Layer and then ran through RISE. The output is given as three separate parts. 1) The token saliency based off of RISE where each word in the phrase is given a color from dark red to bright red where the brighter the red means the token is more NSFW. 2) Concept activations in a bar chart where the different NSFW concepts are on the x axis and the higher value a concept has means that the concept is more present in the phrase. 3) A concept matrix for each token and concept where based on the color it can be seen which tokens contribute to each concept.

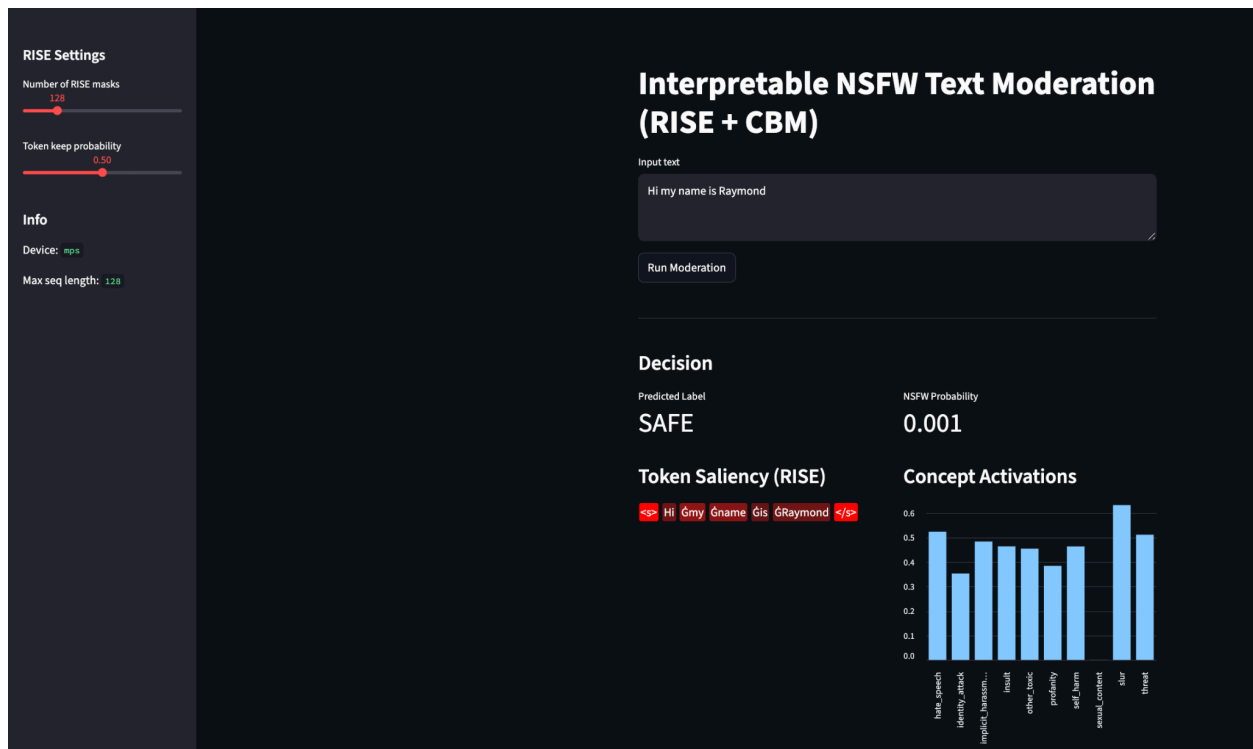


Figure 5: Streamlit visualization of the Interpretable NSFW Text Moderation

We found that our visualization does a good job of highlighting token saliency via the colors on each token (word). However there were some bugs regarding the start and the end of each phrase containing `<s>` and `</s>` which most likely comes from html and the strange character “G” before each token. This is due to the fact that it represents a leading space in the tokenizer. Beyond this the RISE based token saliency and NSFW probability seem to work well.

With regards to the concept activation bar chart and concept matrix we found that it does show the concept activation for each concept, but for some reason the “sexual content” concept seems to always be activated the most for NSFW phrases and activated the least for non NSFW phrases (fig 6). We guess that this is due to the limited training of the concept bottleneck model and possibly due to the training examples mostly being sexual content examples.



Figure 6: One visualization example of NSFW and one example of not NSFW phrases

Unfortunately there was no time for a user study but we hope that our observations would be enough for the scope of this project.

Code can be found at:

<https://github.com/RaymondHKang/ECS289HFinalProject/tree/raymondbranch>

6. Limitations and Future Works

There are many limitations regarding the messiness of the dashboard being the unimportant characters in the RISE token saliency and the concept bottleneck model bar chart mostly only showing large or small values for the “sexual content” example.

We propose two future works:

- 1) Use larger compute resources and a larger variety of datasets to train the RoBERTa (or whatever state of the art text classifier) model as well as the Concept Bottleneck Model
- 2) Apply our approach of the visualization dashboard and RISE/CBM on top of a classifier for multi modal NSFW classification

The first approach for future work would most likely improve accuracy of NSFW detection and the concept bottleneck model.

For the second approach there are lots of examples of NSFW media in videos and photos along with text in social media and many other environments so this would be a great extension of our current work.

Additionally, conducting the user study mentioned in 4.3 would bring great user feedback into both the visualization’s design and allow for a wide variety of testing that would increase the robustness of our model.

7. Conclusion

We introduced a novel way to interpret NSFW text moderation via RISE and Post hoc concept bottleneck models with visualization. This is a topic in which there may be applications in industry (ie in media or work environments) and is important for auditing and legal cases. We apply two new interpretability concepts (RISE and CBM) into a classification model. We converted the concept bottleneck model from image classification to be applied to text classification.

8. Team and Responsibilities

Raymond Kang — Lead researcher and engineer: base model fine-tuning, RISE adaptation, concept head training, and study coordination, dataset curation, annotation protocol, metrics pipeline, and statistical analysis.

Yunhan Wei — Visualization & UX: dashboard design, usability study instruments, and front-end implementation.

9. References

[1] [Petsiuk, Vitali, Abir Das, and Kate Saenko. “RISE: Randomized Input Sampling for Explanation of Black-Box Models.” arXiv preprint arXiv:1806.07421 \(2018\).](#)

[2] [Koh, Pang Wei, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Brunskill, and Percy Liang. “Concept Bottleneck Models.” Proceedings of the 37th International Conference on Machine Learning \(ICML\), 2020.](#)

[3] Koh, Pang Wei, et al. "Concept Bottleneck Models." arXiv preprint arXiv:2007.04612 (2020).

[4] Kim, Been, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viégas, and Rory Sayres. "Interpretability Beyond Feature Attribution: Testing with Concept Activation Vectors (TCAV)." Proceedings of the 35th International Conference on Machine Learning (ICML), 2018.

[5] Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "'Why Should I Trust You?' Explaining the Predictions of Any Classifier." Proceedings of the 22nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2016.

[6] Lundberg, Scott, and Su-In Lee. "A Unified Approach to Interpreting Model Predictions." Advances in Neural Information Processing Systems (NeurIPS), 2017.

[7] Sundararajan, Mukund, Ankur Taly, and Qiqi Yan. "Axiomatic Attribution for Deep Networks." Proceedings of the 34th International Conference on Machine Learning (ICML), 2017.

[8] DeYoung, Jay, et al. "ERASER: A Benchmark to Evaluate Rationalized NLP Models." Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), 2020.

[9] ERASER Project Site. "ERASER Benchmark." <https://www.eraserbenchmark.com/>.

[10] Davidson, Thomas, Dana Warmusley, Michael Macy, and Ingmar Weber. "Automated Hate Speech Detection and the Problem of Offensive Language." Proceedings of ICWSM, 2017.

[11] Jigsaw/Conversation AI. "Jigsaw Unintended Bias in Toxicity Classification." Kaggle Dataset. <https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification>.

[12] Mathew, Binny, Punyajoy Saha, Hiranmayi Reddy, et al. "HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection." Proceedings of the AAAI Conference on Artificial Intelligence, 2021.

[13] HateXplain GitHub Repository. <https://github.com/hate-alert/HateXplain>.

[14] Borkan, Daniel, Lucas Dixon, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. “Nuanced Metrics for Measuring Unintended Bias with Real Data for Text Classification.” Proceedings of the 28th International Conference on World Wide Web Companion, 2019.

[15] Röttger, Paul, Bertram Vidgen, Dirk Hovy, and Janet Pierrehumbert. “HateCheck: Functional Tests for Hate Speech Detection Models.” Proceedings of the 59th Annual Meeting of the ACL, 2021.

[16] HateCheck GitHub Repository. <https://github.com/paul-rottger/hatecheck-data>.

[17] Liu, Yinhan, Myle Ott, Naman Goyal, et al. “RoBERTa: A Robustly Optimized BERT Pretraining Approach.” arXiv preprint arXiv:1907.11692 (2019).

[18] Sanh, Victor, Lysandre Debut, Julien Chaumond, and Thomas Wolf. “DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter.” arXiv preprint arXiv:1910.01108 (2019).

[19] Guo, Chuan, et al. “On Calibration of Modern Neural Networks.” Proceedings of the 34th International Conference on Machine Learning (ICML), 2017.

[20] PyTorch Captum Documentation. <https://captum.ai/>.

[21] Hugging Face Transformers Documentation. <https://huggingface.co/docs/transformers>.

[22] Jain, Sarthak, and Byron C. Wallace. “Attention Is Not Explanation.” Proceedings of EMNLP, 2019.

[23] Wiegreffe, Sarah, and Yuval Pinter. “Attention Is Not Not Explanation.” Proceedings of EMNLP, 2019.